JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON MONDAY 06th APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

1. Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \sin^{-1}\left(\frac{x+[x]}{3}\right)$ is $[\alpha, \beta)$, then $\alpha^2 + \beta^2$ is equal to:

(1) 2 (2) 5

(3) 10 (4) 13

Sol. $-1 \le \frac{x+[x]}{3} \le 1$

$\Rightarrow -3 \le x - 1[x] \le 3$

$\Rightarrow -3 - [x] \le x \le 3 - [x]$

if $[x] = -1 \Rightarrow x \in [-1,0)$

if $[x] = 0 \Rightarrow x \in [0,1)$

if $[x] = 1 \Rightarrow x \in [1,2)$

Hence $x \in [-1,2)$

$\Rightarrow \alpha = -1$ and $\beta = 2$

$\Rightarrow \alpha^2 + \beta^2 = 5$

2. Let one root of the quadratic equation in x:

$(k^2 - 15k + 27)x^2 + 9(k-1)x + 18 = 0$

be twice the other. Then the length of the latus rectum of the parabola $y^2 = 6kx$ is equal to:

(1) 4 (2) 6

(3) 8 (4) 12

Sol. $\alpha + 2\alpha = \frac{-9(k-1)}{k^2 - 15k + 27} = 3\alpha \quad \ldots(1)$

$\alpha \times 2\alpha = \frac{18}{k^2 - 15k + 27} = 2\alpha^2 \quad \ldots(2)$

solving (1) & (2)

$k = 2$

$\Rightarrow$ parabola will be $y^2 = 6kx = 12x$

length of latus rectum = 12

3. Let e_1 and e_2 be two distinct roots of the equation $x^2 - ax + 2 = 0$. Let the sets

$\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of hyperbolas$\} = (\alpha, \beta)$, and $\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of an ellipse and a hyperbola, respectively$\} = (\gamma, \infty)$.

The $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

(1) 18 (2) 22

(3) 26 (4) 34

$e_1 + e_2 = a$

$e_1 e_2 = 2 \quad \Rightarrow e_2 = \frac{2}{e_1}$

Case 1 : Both are eccentricities of hyperbola

Now, $e_1 > 1$ and $e_2 > 1$

$\Rightarrow \frac{2}{e_1} > 1 \Rightarrow e_1 < 2 \quad \Rightarrow e_1 \in (1,2)$

Now, $a = e_1 + \frac{1}{e_1}$

Minimum occurs at $e_1 = \sqrt{2}$

$\Rightarrow a_{min} = 2\sqrt{2}$

if e = 1,2 (end points)

$a \to 2,3 \quad \Rightarrow a \in \left(2\sqrt{2}, 3\right)$

$\Rightarrow \alpha = 2\sqrt{2}$ and $\beta = 3$

Case 2 : One ellipse , one hyperbola

(e_1) (e_2)

$0 < e_1 < 1$ and $e_2 < 1$

$e_2 = \frac{2}{e_1} > 2$

Now $a = e_1 + \frac{2}{e_1}$

as $e_1 \to 1 \Rightarrow a \to 3$

$e_1 \to 0 \Rightarrow a \to \infty$

$\therefore (3,\infty) \Rightarrow r = 3$

Now $\alpha^2 + \beta^2 + \gamma^2 = 8 + 9 + 9 = 26$

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4. Let the set of all values of $k \in R$ such that the equation $z(\bar{z}+2+i)+k(2+3i)=0$, $z \in C$, has at least one solution, be the interval $[\alpha, \beta]$. Then $9(\alpha+\beta)$ is equal to:

(1) -10 (2) -8

(3) $10\sqrt{13}$ (4) $8\sqrt{13}$

Sol. Put $z = x + iy$, $\bar{z} = x - iy$

$\Rightarrow \bar{z}+2+i = x-iy+2+i$

$\Rightarrow z(\bar{z}+2+i) = (x+iy)(x+2+i(1-y))$

$\Rightarrow$ Real part $= x(x+2) - y(1-y)$

Imaginary part $= x(1-y) + y(x+2)$

$\Rightarrow x^2 + y^2 + 2x - y + 2k = 0$ ..(1)

and $x + 2y + 3k = 0$ …(2)

eliminating x from (1) and (2)

$5y^2 + (12k - 5)y + 9k^2 - 4k = 0$

since $y \in R$

use $D \geq 0$

$(12k-5)^2 - 4.5(9k^2 - 4k) \geq 0$

$\Rightarrow 36k^2 + 40k - 25 \leq 0$

$\Rightarrow \alpha+\beta = \frac{-10}{9}$

$\Rightarrow 9(\alpha+\beta) = -10$

5. The value of $1^3 - 2^3 + 3^3 - .... + 15^3$ is:

(1) 1706 (2) 1856

(3) 1982 (4) 2403

Sol. $(1^3 - 2^3) + (3^3 - 4^3) + ... 13^3 - 14^3 + 15^3$

use $n^3 - (n+1)^3 = -3n^2 - 3n - 1$

put n = 1,3,5,….13 (7 terms)

sum of pairs = – 1519

add last term = 15^3

$\Rightarrow 3375 - 1519 = 1856$

6. The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8. If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:

(1) $\frac{34}{9}$ (2) $\frac{34}{13}$

(3) $\frac{32}{9}$ (4) $\frac{32}{13}$

Sol. Sum of 10 terms of A.P, $S_{10} = 160$

$5[2a + 9d] = 160$

$2a + 9d = 32$

G.P. $\Rightarrow$ d, da, da^2….

$\Rightarrow d + da = 8$

$\Rightarrow \frac{d + d(32 - 9d)}{2} = 8$

$\Rightarrow 2d + 32d - 9d^2 = 8$

$\Rightarrow 9d^2 - 34d + 8 = 0$

$d_1 + d_2 = \frac{34}{9}$

7. The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is:

(1) 2670 (2) 2840

(3) 2920 (4) 3600

Sol. INCONSEQUENTIAL contains

(I, E,O, A,U) and (N,C,S,Q,T,L)

Total number of 4 letter words using two vowels and two consonants is

$= {}^5C_2 \times {}^6C_2 \times 4!$

$= 3600$

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8. If the coefficients of the middle terms in the binomial expansions of $(1 + \alpha x)^{26}$ and $(1 - \alpha x)^{28}$, $\alpha \neq 0$, are equal, then the value of α is:

(1) 1 (2) $\frac{14}{13}$

(3) $\frac{27}{7}$ (4) $\frac{7}{27}$

Sol. $(1 + \alpha x)^{26}$

Middle term, $T_{14} = {}^{26}C_{13}\alpha^{13}$

For $(1 - \alpha x)^{28}$

Middle term $= {}^{28}C_{14}\alpha^{14}$

$\Rightarrow {}^{26}C_{13}\alpha^{13} = {}^{28}C_{14}\alpha^{14}$

$\Rightarrow {}^{26}C_{13} = {}^{28}C_{14}\alpha$

$\Rightarrow {}^{26}C_{13} = \frac{28}{14} \times \frac{27}{14} \cdot {}^{26}C_{13}\alpha$

$\Rightarrow \alpha = \frac{7}{27}$

9. A data consists of 20 observations $x_1, x_2, ..., x_{20}$. If $\sum_{i=1}^{20}(x_i + 5)^2 = 2500$ and $\sum_{i=1}^{20}(x_i - 5)^2 = 100$, then the ratio of mean to standard deviation of this data is :

(1) 2 : 1 (2) 3 : 1

(3) 3 : 2 (4) 4 : 1

Sol. $\sum_{i=1}^{20}(x_i + 5)^2 = 2500$ ...(1)

$\sum_{i=1}^{20}(x_i - 5)^2 = 100$ ...(2)

on (1) - (2) $\Rightarrow 20\sum x_i = 2400 \Rightarrow \sum x_i = 120$

mean, $\bar{x} = 6$

(1) + (2) $\Rightarrow 2\sum x_i^2 + 40(5^2) = 2600$

$\sum x_i^2 = 800$

$\sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2 = \frac{800}{20} - 36 = 4$

$\bar{x} : \sigma = 6 : 2 = 3 : 1$

10. A bag contains (N + 1) coins – N fair coins, and one coin with 'Head' on both sides. A coin is selected at random and tossed. If the probability of getting 'Head' is $\frac{9}{16}$, then N is equal to:

(1) 5 (2) 7

(3) 8 (4) 9

Sol. Total (N + 1) coins

$N \Rightarrow$ fair coins

$1 \Rightarrow$ Head on both side

$P(1^+) = \left(\frac{N}{N+1}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{N+1}\right)1$

$= \frac{N+2}{2(N+1)} = \frac{9}{16}$

$8N + 16 = 9N + 9$

$\Rightarrow N = 7$

11. If the eccentricity e of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, passing through $(6, 4\sqrt{3})$, satisfies $15(e^2 + 1) = 34e$, then the length of the latus rectum of the hyperbola $\frac{x^2}{b^2} - \frac{y^2}{2(a^2+1)} = 1$ is:

(1) 10 (2) 20

(3) 25 (4) 30

Sol. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

It passs through $(6, 4\sqrt{3})$

$\Rightarrow \frac{36}{a^2} - \frac{48}{b^2} = 1$ ...(1)

also $15e^2 - 34e + 15 = 0$

$\Rightarrow 15e^2 - 25e - 9e + 15 = 0$

$e = \frac{5}{3}$ or $\frac{3}{5} \Rightarrow e = \frac{5}{3}$

$1 + \frac{b^2}{a^2} = \frac{25}{9} \Rightarrow \frac{b^2}{a^2} = \frac{16}{9}$ ...(2)

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using (1) and (2)

$\Rightarrow \frac{36}{a^2}-\frac{48}{16a^2}\times 9=1 \Rightarrow a=3, b=4$

Length of L.R.

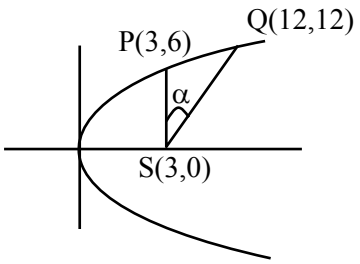
$\frac{4\left(a^2+1\right)}{b}=\frac{4(10)}{4}=10$

12. Let chord PQ of length $3\sqrt{13}$ of the parabola $y^2 = 12x$ be such that the ordinates of points P and Q are in the ratio 1 : 2. If the chord PQ subtends an angle α at the focus of the parabola, then sin α is equal to :

(1) $\frac{3}{5}$ (2) $\frac{4}{5}$

(3) $\frac{5}{13}$ (4) $\frac{12}{13}$

Sol.



$P(3t^2, 6t)$, $Q(12t^2, 12t)$

$PQ=\sqrt{81t^4+36t^2}$

$81t^4 + 36t^2 = 117$

$\Rightarrow 9t^4 + 4t^2 - 13 = 0$

$(t^2 - 1)(9t^2 + 13) = 0 \Rightarrow t = 1$

P(3,6), Q(12,12)

slope of QS $= \frac{4}{3}$

$\tan\alpha=\frac{3}{4}, \sin\alpha=\frac{3}{5}$

13. Let $0 < \alpha < 1$, $\beta = \frac{1}{3\alpha}$ and $\tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta) = \frac{\pi}{4}$. Then $6(\alpha + \beta)$ is equal to:

(1) 6 (2) 7

(3) 8 (4) 9

Sol. $\beta=\frac{1}{3\alpha}, \tan^{-1}(1-\alpha)+\tan^{-1}(1-\beta)=\frac{\pi}{4}$

$\Rightarrow \tan\left(\tan^{-1}(1-\alpha)+\tan^{-1}(1-\beta)\right)=1$

$\Rightarrow \frac{1-\alpha+1-\beta}{1-(1-\alpha)(1-\beta)}=1$

$\Rightarrow 2-\alpha-\beta=1-(1-\alpha-\beta+\alpha\beta)$

$\Rightarrow 2-\alpha-\beta=\alpha+\beta-\alpha\beta$

$\Rightarrow 2(\alpha+\beta)=2+\alpha\beta$

$\Rightarrow 2(\alpha+\beta)=2+\frac{1}{3}=\frac{7}{3}$

$\alpha+\beta=\frac{7}{6}$

$6(\alpha + \beta) = 7$

14. Let $S=\left\{\theta\in(-2\pi, 2\pi): \cos\theta+1=\sqrt{3}\sin\theta\right\}$.

Then $\sum_{\theta\in S}\theta$ is equal to:

(1) $-\frac{2\pi}{3}$ (2) $-\frac{4\pi}{3}$

(3) $\frac{2\pi}{3}$ (4) $\frac{4\pi}{3}$

Sol. $\cos\theta + 1 = \sqrt{3}\sin\theta$

$\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}+1=\sqrt{3}\left(\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right)$

$2=2\sqrt{3}\tan\frac{\theta}{2}$

$\Rightarrow \tan\frac{\theta}{2}=\frac{1}{\sqrt{3}}$ $\quad \theta\in(-2\pi, 2\pi), \ \frac{\theta}{2}\in(-\pi,\pi)$

$\frac{\theta}{2}=-\frac{5\pi}{6}, \frac{\pi}{6}$

$\theta=\frac{-5\pi}{3}, \frac{\pi}{3}$

Sum $=-\frac{5\pi}{3}+\frac{\pi}{3}=\frac{-4\pi}{3}$

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15. Let the image of the point P(1, 6, a) in the line $L: \frac{x}{1}=\frac{y-1}{2}=\frac{z-a+1}{b}, b>0$, be $\left(\frac{a}{3}, 0, a+c\right)$.

If $S(\alpha, \beta, \gamma)$, $\alpha > 0$, is the point on L such that the distance of S from the foot of perpendicular from the point P on L is $2\sqrt{14}$, then $\alpha + \beta + \gamma$ is equal to :

(1) 19 (2) 20

(3) 21 (4) 22

Sol. Let Q be the point $\perp^r$ from P(1,6,a) to the line

$$L: \frac{x}{1}=\frac{y-1}{2}=\frac{z-a+1}{b}$$

The image of point P in line L is $P'\left(\frac{a}{3}, 0, a+c\right)$

$\because$ Q is the mid point PP' :

$$Q \equiv \left(\frac{1+\frac{a}{3}}{2}, \frac{6+0}{2}, \frac{a+a+c}{2}\right) \equiv \left(\frac{3+a}{6}, 3, \frac{2a+c}{2}\right)$$

comparing x and y, $\frac{3+a}{6}=1 \Rightarrow a=3$

$\therefore\ Q \equiv \left(1, 3, \frac{6+c}{2}\right)$

Direction vector of L is $\vec{v}=(1,2,b)$

vector PQ is $\perp^r$ to $\vec{v}$

$\therefore\ \overrightarrow{PQ}=\left(1-1, 3-6, \frac{6+c}{2}-3\right)$

$=\left(0, -3, \frac{c}{2}\right)$

$\overrightarrow{PQ}.\vec{v}=0$

$(0).(1)+(-3)(2)+\left(\frac{c}{2}\right)(b)=0$

$bc = 12$

Now comparing z with $Q\left(1, 3, \frac{6+c}{2}\right)$

$$\frac{3-1}{2}=\frac{\frac{6+c}{2}-3+1}{b}$$

$2b - c = 2$

solving $bc = 12$ and $c = 2b - 2$

$b = 3, -2$

$b = 3 \quad (b > 0)$

$c = 4$

$\therefore$ Q(1,3,5)

Now, any point S on the line L can be written as (k, 2k + 1, 3k + 2)

$$SQ=\sqrt{(k-1)^2+(2k-2)^2+(3k-3)^2}=2\sqrt{14}$$

$k = 3, -1$

for $k = 3$, $S \equiv (3,7,11)$

$\therefore\ \alpha+\beta+\gamma=21$

16. Let a line L be perpendicular to both the lines

$$L_1: \frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7} \text{ and}$$

$$L_2: \frac{x-2}{1}=\frac{y-4}{4}=\frac{z-6}{7}.$$

If θ is the acute angle between the lines L and

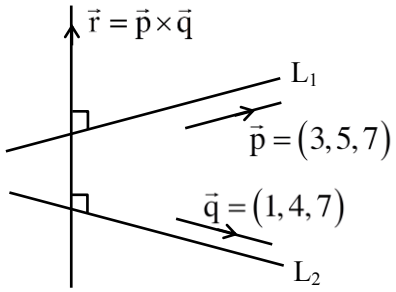
$L_3: \frac{x-\frac{8}{7}}{2}=\frac{y-\frac{4}{7}}{1}=\frac{z}{2}$, then $\tan\theta$ is equal to:

(1) $\frac{3}{2}\sqrt{2}$ (2) $\frac{5}{2}\sqrt{2}$

(3) $\frac{5}{3}\sqrt{2}$ (4) $\frac{4}{3}\sqrt{2}$

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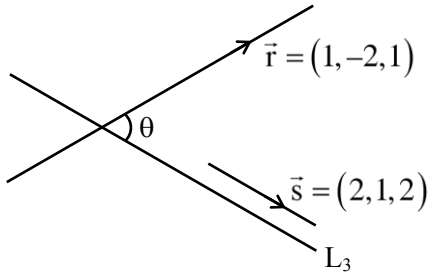
Sol.



$$\vec{r}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 5 & 7\\ 1 & 4 & 7\end{vmatrix}=7\hat{i}-14\hat{j}+7\hat{k}$$

$$\equiv 7\left(\hat{i}-2\hat{j}+\hat{k}\right)$$

Now,



$$\because \cos\theta=\frac{2-2+2}{\sqrt{1+4+1}\sqrt{4+1+4}}$$

$$\Rightarrow \cos\theta=\frac{2}{3\sqrt{6}}\Rightarrow \tan\theta=\frac{5\sqrt{2}}{2}$$

17. The value of $\lim\limits_{x\to 0}\left(\dfrac{x^2\sin^2 x}{x^2-\sin^2 x}\right)$ is:

(1) 2 (2) 3

(3) 4 (4) 6

Sol. $\lim\limits_{x\to 0}\dfrac{x^2\sin^2 x}{x^2-\sin^2 x}$

Applying expansion

$$\lim_{x\to 0}\frac{x^2\sin^2 x}{x^2-\left(x-\frac{x^3}{3!}+\ldots\right)^2}$$

$$\lim_{x\to 0}\frac{x^2\sin^2 x}{x^2-x^2+\frac{2x^4}{3!}\ldots}=\frac{3!}{2}=3$$

18. The value of the integral $\int\limits_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\dfrac{32\cos^4 x}{1+e^{\sin x}}\right)dx$ is :

(1) $4\pi+2$ (2) $3\pi+8$

(3) $3\pi+4$ (4) $4\pi+3$

Sol. Apply (P-5)

$$I=\int_{-\pi/4}^{\pi/4}\frac{32\cos^4\theta}{1+e^{-\sin\theta}}d\theta$$

$$\text{Add } 2I=\int_{-\pi/4}^{\pi/4}32\cos^4\theta\,d\theta=2\int_0^{\pi/4}32\cos^4\theta\,d\theta$$

$$I=32\int_0^{\pi/4}\cos^4\theta\,d\theta$$

$$=8\int_0^{\pi/4}(2\cos^2\theta)^2d\theta=8\int_0^{\pi/4}(1+\cos2\theta)^2d\theta$$

$$=8\int_0^{\pi/4}1+2\cos2\theta+\cos^2 2\theta\,d\theta$$

$$=8\int_0^{\pi/4}1+2\cos2\theta+\frac{1+\cos4\theta}{2}d\theta$$

$$=8\left[\frac{3\theta}{2}+\frac{2\sin2\theta}{2}+\frac{\sin4\theta}{8}\right]_0^{\pi/4}$$

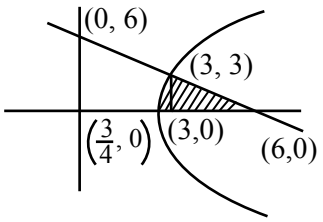
$$=8\left[\left(\frac{3\pi}{8}+1+0\right)\right]=3\pi+8$$

19. The area of the region $\{(x, y) : 0 \le y \le 6 - x, y^2 \ge 4x - 3, x \ge 0\}$ is :

(1) 8 (2) 9

(3) 12 (4) 15

Sol. $A=\int_0^3\left(6-y-\dfrac{y^2+3}{4}\right)dy=9$



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20. Let e be the base of natural logarithm and let $f: \{1, 2, 3, 4\} \to \{1, e, e^2, e^3\}$ and $g : \{1, e, e^2, e^3\} \to \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\right\}$ be two bijective functions such that f is strictly decreasing and g is strictly increasing. If $\phi(x) = \left[f^{-1}\left\{g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x$, then the area of the region $R = \{(x, y) : x^2 \le y \le \phi(x), 0 \le x \le 1\}$ is :

(1) $\frac{3-\log_e(2)}{3\log_e(2)}$ (2) $\frac{1}{3\log_e(2)}$

(3) $3 + \log_e(2)$ (4) $\frac{3+\log_e(2)}{2+\log_e(3)}$

Sol. $f(x) = e^{x-1}$

$g(x) = \frac{1}{\log_e x + 1}$

$f^{-1}(g^{-1}(1/2)) = f^{-1}(e) = 2$

$\phi(x) = 2^x$

$\therefore \int_0^1 (2^x - x^2)dx = \left(\frac{2^x}{\ln 2} - \frac{x^3}{3}\right)_0^1 = \frac{1}{\ln 2} - \frac{1}{3}$

$= \frac{3 - \ell n 2}{3 \ell n 2}$

SECTION-B

21. Let $A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ satisfy

$A^2 + \alpha(\text{adj (adj (A)))} + \beta(\text{adj (A) (adj (adj (A))))}$

$= \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ for

Some $\alpha, \beta \in \mathbb{R}$.

Then $(\alpha - \beta)^2$ is equal to _____

Sol. $|A| = -1$

$\text{adj}(\text{adj}(A)) = |A|^{n-2}A = |A|^{3-2}.A = |A|.A$

$\text{adj}(A)\ (\text{adj}\ (\text{adj}(A))) = \text{adj}(A)\ (|A|.A)$

$= |A|\ (\text{adj}(A).A)$

since $\text{adj}(A)A = |A|I$,

then $\text{adj}(A)\ (\text{adj}(\text{adj}(A))) = |A|^2 I$

or $\text{adj}(A)(\text{adj}\ (\text{adj}\ (A))) = I$

solving the matrix equation

$$A^2 + \alpha(-A) + \beta(I) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

$A^2 - \alpha A + \beta I = M$

$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \alpha \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} + \beta \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$\alpha = 1$ and $\beta = -1$

$\therefore (\alpha - \beta)^2 = 4$

22. Let the centre of the circle $x^2 + y^2 + 2gx + 2fy + 25 = 0$ be in the first quadrant and lie on the line $2x - y = 4$. Let the area of an equilateral triangle inscribed in the circle be $27\sqrt{3}$. Then the square of the length of the chord of the circle on the line $x = 1$ is _____.

Sol. $x^2 + y^2 + 2gx + 2fy + 25 = 0$

$C \equiv (-g, -f)$ lies on line

$\therefore -2g + f = 4$ ...(i)

Also $g^2 + f^2 - 25 = r^2$ ...(ii)

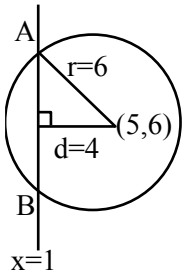
Area of equation triangle inside the circle $= \frac{3\sqrt{3}}{4} r^2$

$27\sqrt{3} = \frac{3\sqrt{3}}{4} \times r^2$

$r^2 = 36$

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Solving (i) and (ii) g = –5, f = –6



$\ell_{AB} = \sqrt{36-16} = 2\sqrt{20}$

$(\ell_{AB})^2 = 80$

23. If $\vec{a} = \hat{i}+\hat{j}+\hat{k}, \vec{b} = \hat{j}-\hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a}\times\vec{c} = \vec{b}$ and $\vec{a}\cdot\vec{c} = 3$, then $\vec{c}\cdot(\vec{a}-2\vec{b})$ is equal to _____.

Sol. Using triple product identity with $\vec{a}\times\vec{c} = \vec{b}$

$\vec{a}\times(\vec{a}\times\vec{c}) = \vec{a}\times\vec{b}$

$(\vec{a}.\vec{c})\vec{a} - (\vec{a}.\vec{a})\vec{c} = \vec{a}\times\vec{b}$

As $\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix} = -2\hat{i}+\hat{j}+\hat{k}$

Substituting given values

$3\vec{a} - 3\vec{c} = -2\hat{i}+j+k$

$3(\hat{i}+\hat{j}+\hat{k}) + 2\hat{i}+\hat{j}+\hat{k} = 3\vec{c}$

$\vec{c} = \frac{1}{3}(5\hat{i}+2\hat{j}+2\hat{k})$

Now $\vec{c}.(\vec{a}-2\vec{b}) = \frac{1}{3}(5\hat{i}+2\hat{j}+2\hat{k}).(\hat{i}-\hat{j}+3\hat{k})$

$= \frac{1}{3}[5-2+6] = 3$

24. For the functions $f(\theta) = \alpha\tan^2\theta + \beta\cot^2\theta$, and $g(\theta) = \alpha\sin^2\theta + \beta\cos^2\theta$, $\alpha > \beta > 0$, let $\min\limits_{0<\theta<\frac{\pi}{2}} f(\theta) = \max\limits_{0<\theta<\pi} g(\theta)$. If the first term of a G.P. is $\left(\frac{\alpha}{2\beta}\right)$, its common ratio is $\left(\frac{2\beta}{\alpha}\right)$ and the sum of its first 10 terms is $\frac{m}{n}$, gcd(m, n) = 1, then m + n is equal to _____.

Sol. $f(\theta)\big|_{min} = 2\sqrt{\alpha\beta}$ (AM ≥ GM)

$g(\theta) = \alpha\sin^2\theta + \beta(1-\sin^2\theta) = (\alpha-\beta)\sin^2\theta + \beta$

$\therefore\ g(\theta)\big|_{max} = \alpha$

Equation $2\sqrt{\alpha\beta} = \alpha$

$4\alpha\beta = \alpha^2$

$\alpha = 4\beta\ (\alpha \neq 0)$

Now first term $\left(\frac{\alpha}{2\beta}\right) = 2$ and common ratio $\left(\frac{2\beta}{\alpha}\right) = \frac{1}{2}$

$S_{10} = \frac{a(1-r^{10})}{1-r} = \frac{1023}{256} = \frac{m}{n}$

m + n = 1279

25. Let y = y (x) be the solution of the differential equation $\left(x^2 - x\sqrt{x^2-1}\right)dy + \left(y\left(x-\sqrt{x^2-1}\right)-x\right)dx = 0$, $x \geq 1$. If y(1) = 1, then the greatest integer less than $y(\sqrt{5})$ is _____.

Sol. $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x-\sqrt{x^2-1}}$

$\frac{dy}{dx} + \frac{y}{x} = x + \sqrt{x^2-1}$

I.f $= e^{\ln x} = x$

$y.x = \int\left(x+\sqrt{x^2-1}\right)x\,dx$

$= \frac{x^2}{2} + \frac{(x^2-1)^{3/2}}{3} + C$

Given y(1) = 1

$1 = \frac{1}{2} + C \Rightarrow C = \frac{1}{2}$

$\left[y(\sqrt{5})\right] = \left[\frac{\sqrt{5}}{2} + \frac{8}{3} + \frac{1}{2\sqrt{5}}\right] = 3$

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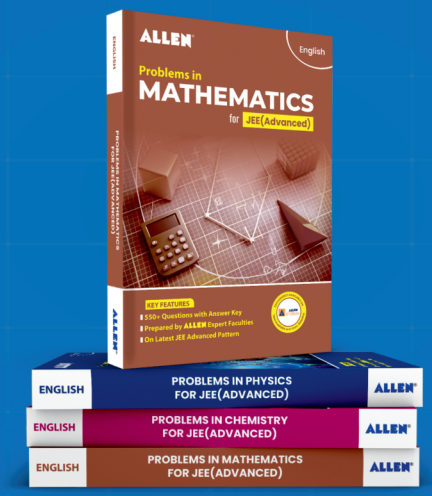


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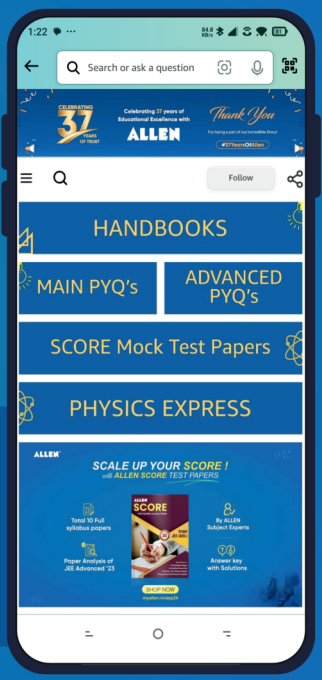
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